
Federated Conformal Prediction for Mitigating Hallucinations in Federated Fine-Tuned Vision-Language Models via Abstention

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Abstract

Vision-Language Models (VLMs) fine-tuned in federated settings allow privacy-preserving multimodal reasoning across distributed clients. However, they hallucinate because the model commits to a wrong answer with high confidence. In our setup without any safety mechanism, a federated LoRA fine-tuned VLM hallucinated on 13.05% of all test examples across five non-IID clients.

We propose **Federated Binned Local-External Nonconformity Distribution (Fed-BLEND)**, a post-hoc federated conformal prediction framework that reduces hallucinations by abstaining, with a formal coverage guarantee of at least 90%. Fed-BLEND bins each client’s calibration examples by predictive entropy into easy and hard strata, then shrinks the local per-bin quantile toward a global pooled quantile in proportion to local sample size. This produces tighter, client-specific prediction sets without sharing raw data.

Applied to a federated LoRA fine-tuned Qwen/Qwen2.5-VL-3B-Instruct model on ScienceQA across five heterogeneous clients, Fed-BLEND cuts hallucinations from 13.05% to 3.79% a 71% relative reduction while having a useful-answer rate of 76.63% and Precision@Commit of 95.29%. It achieves the best safety and efficiency trade-off among all evaluated baselines, including Global-LAC, Local-LAC, Local-APS, and Fed-DSCP.

1 Introduction

Vision-Language Models (VLMs) process text and images jointly and achieve high accuracy on multimodal benchmarks. Supervised fine-tuning (SFT) with LoRA [Hu et al., 2022] [Bohnet et al., 2025] adapters is the standard way to adapt a pretrained VLM to a domain-specific dataset. In privacy-sensitive domains- like medical imaging, legal analysis, educational platforms data cannot be moved to a central server. Federated learning (FL) [McMahan et al., 2017] lets multiple clients train a shared model without exchanging raw data.

Even after federated fine-tuning, VLMs hallucinate. Hallucination here means the model picks one answer and commits to it, even when that answer is wrong. This is different from ordinary classification error where the model gives no signal of uncertainty, so the downstream system has no reason to distrust the output. In high-stakes settings, such silent errors are costly.

Table 2 makes this clear. A federated LoRA-tuned Qwen/Qwen2.5-VL-3B-Instruct model using flower framework [Beutel et al., 2020] reaches 86.95% top-1 accuracy on ScienceQA [Lu et al., 2022] yet hallucinates on 13.05% of test examples on non-iid data. Client 1 alone hits 25.27%.

Our contribution. We propose **Fed-BLEND** (Federated Binned Local-External Nonconformity Distribution), a post-hoc framework that wraps any federated Language model or VLM with conformal prediction to suppress hallucinations via abstention. Instead of always committing to a single answer, the model outputs a *prediction set* which means a subset of candidate answers that contains the correct answer with probability at least $1 - \alpha$. When the set has more than one element, the model abstains rather than risk a wrong commit.

Fed-BLEND makes three contributions:

1. **Entropy-based difficulty binning.** Each client splits its calibration examples into easy and hard bins using the median predictive entropy computed on a local validation set. A separate conformal threshold is fitted per bin, giving tighter sets on confident examples and wider sets on uncertain ones.
2. **Sample-size-aware shrinkage.** Each client’s local per-bin quantile is shrunk toward a global pooled quantile by the weight $\lambda = n/(n + \tau)$, where n is the local bin count and τ is a tuned constant. When n is small the global estimate dominates; when n is large the local estimate dominates. This keeps sparse bins from producing unreliable thresholds.
3. **Formal coverage guarantee.** Under exchangeability within each client, Fed-BLEND guarantees marginal coverage $\Pr(y^* \in \mathcal{C}(x)) \geq 1 - \alpha$ for $\alpha = 0.10$.

On a five-client non-IID ScienceQA benchmark, Fed-BLEND reduces hallucinations from 13.05% to 3.79% while meeting the 90% coverage target and keeping the useful-answer rate at 76.63%.

2 Related Work

Conformal prediction. Recent hallucination mitigation systems use a *verification loop* instead of trusting the model’s internal memory. A common approach is Retrieval-Augmented Generation (RAG), where the model grounds answers in retrieved documents; newer variants add extra checking steps when retrieval looks unreliable [Ayala and Bechard, 2024]. Corrective RAG (CRAG) uses a retrieval evaluator to decide when to trigger corrective actions during generation [Yan et al., 2024]. In production, guardrail frameworks such as NVIDIA NeMo Guardrails enforce grounding and run fact-checking during inference [Rebedea et al., 2023].

For large language models (LLMs), conformal methods have evolved from “predicting a label” to “generating text.” Conformal Language Modeling constructs sets of possible text generations using principled stopping rules that provide statistical guarantees suited to open-ended generation [Quach et al., 2024]. Related work uses these techniques to help the model abstain when it cannot provide a reliable answer, while bounding the probability of a false claim based on the chosen nonconformity measure [Yadkori et al., 2024].

Our work improves these pipelines by providing a principled uncertainty-handling layer we use *federated conformal prediction* to decide when to verify, replacing ad-hoc thresholds with well-calibrated coverage sets.

Conformal prediction (CP) gives distribution-free, finite-sample coverage guarantees under exchangeability [Shafer and Vovk, 2008, Angelopoulos and Bates, 2023]. In the split-conformal variant, a held-out calibration set yields nonconformity scores whose empirical quantile sets the coverage level. Sadinle et al. [2019] introduced LAC, which minimises expected set size subject to a coverage constraint. Romano et al. [2020] proposed APS, a cumulative-score rule that gives approximate conditional coverage when LAC’s assumptions break. Fontana et al. [2023] survey the field broadly.

Federated conformal prediction. Standard CP requires exchangeability across calibration and test data, an assumption that breaks in federated learning when clients have heterogeneous data distributions. Lu et al. [2023] replace full exchangeability with *partial* exchangeability—data need only be exchangeable within each client—and prove finite-sample coverage in this weaker regime (FCP). Humbert et al. [2023] propose FedCP-QQ, which aggregates local quantiles via a quantile-of-quantiles rule in one communication round. Plassier et al. [2023] handle label shift by reweighting nonconformity scores with importance weights. Nguyen et al. [2026] average local quantiles weighted by calibration-set size (FedWQ-CP), and Xu et al. [2025] use generative density models for approximate conditional coverage (Fed-CCP).

Entropy-aware conformal prediction. Luo and Colombo [2024] reweight the calibration quantile by predictive entropy (ERCC): high-confidence inputs receive smaller prediction sets while low-confidence inputs receive larger ones. They show that predictive entropy is a reliable signal of per-instance conformal difficulty, a finding that directly supports our choice of entropy as the stratification signal.

Conformal abstention for language models. Abbasi-Yadkori et al. [2024] apply CP to selective prediction in LLMs. The model draws multiple output samples, measures their consistency, and abstains when the CP threshold indicates uncertainty is too high. This work directly links conformal abstention to hallucination control in language models.

Where Fed-BLEND fits. No existing federated CP method does all three of the following at once: (i) stratify calibration by difficulty, (ii) shrink per-stratum thresholds by local sample size, and (iii) do both per client rather than globally.

3 Problem Setup

Federated fine-tuning. There are K clients. Client k holds a private dataset $\mathcal{D}_k = \{(x_i, y_i)\}$ of image–question–answer triples drawn from a local, non-IID distribution. Each client fine-tunes a shared pretrained VLM f_θ with LoRA adapters on its training split, and the adapter weights are aggregated via FedAvg to produce a global model $f_{\hat{\theta}}$.

Nonconformity scores. For a multiple-choice question (x, Y) with $|Y| = D$ options, the model returns logits $z_y \in \mathbb{R}$ for each $y \in Y$. Softmax probabilities are

$$p_y = \frac{e^{z_y}}{\sum_{j \in Y} e^{z_j}}. \quad (1)$$

The nonconformity score for the true answer y^* is

$$s(x, y^*) = 1 - p_{y^*}. \quad (2)$$

A small score means the model is confident and correct.

Coverage goal. We want a set-valued predictor $\mathcal{C} : \mathcal{X} \rightarrow 2^Y$ satisfying

$$\Pr(y^* \in \mathcal{C}(x)) \geq 1 - \alpha, \quad \alpha = 0.10. \quad (3)$$

Subject to this constraint, we want to maximise the *useful rate* (fraction of examples where $|\mathcal{C}(x)| = 1$ and the answer is correct) and minimise the *hallucination rate* (fraction where $|\mathcal{C}(x)| = 1$ and the answer is wrong).

4 Data

Dataset. We use **ScienceQA**, a multimodal multiple-choice benchmark covering science, geography, history, and social studies. Each example has an image, a question, four answer choices (A–D), and a correct label.

Non-IID federated split. We split ScienceQA across $K = 5$ clients by subject category. This creates a strongly non-IID partition (Figure 1). Client 2 gets the most training data (1,243 examples); Client 3 gets the least (416). Within each client the data is split four ways (Table 1).

The **validation** split sets each client’s easy/hard entropy threshold. The **calibration** split fits the conformal quantiles $\hat{q}$. The **test** split evaluates all reported metrics. The validation split also serves for the grid search over τ (Section 5.4).

5 Method: Fed-BLEND

Fed-BLEND runs in four stages: (1) federated fine-tuning, (2) entropy binning, (3) local quantile estimation, and (4) shrinkage toward the global quantile. Algorithm 1 gives the full procedure.

Table 1: Data split across five federated clients.

Client	Train	Validation	Calibration	Test
Client 0	511	73	73	74
Client 1	637	91	91	91
Client 2	1,243	178	178	178
Client 3	416	59	60	60
Client 4	504	72	72	72
Overall	3,311	473	474	475

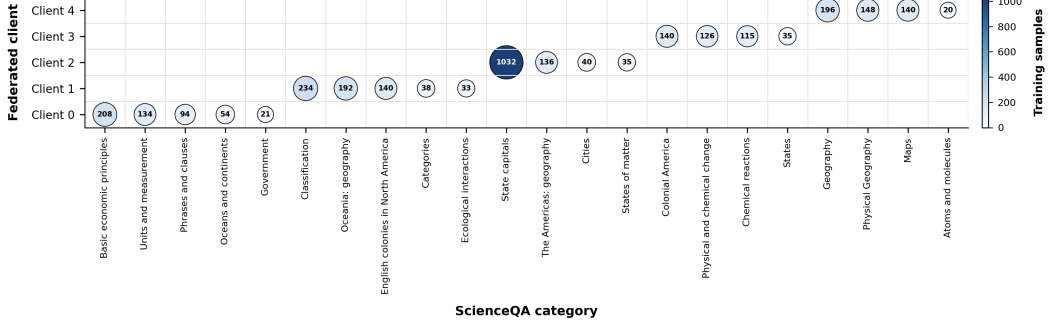


Figure 1: Non-IID data distribution across five federated clients. Each bubble represents one (client, category) pair; bubble area is proportional to training sample count.

5.1 Stage 1: Federated LoRA Fine-Tuning

Client c fine-tunes f_θ on $\mathcal{D}_c^{\text{train}}$ using LoRA adapters ΔW . FedAvg aggregates adapters across rounds:

$$\Delta W^{(t+1)} = \sum_{c=1}^K \frac{n_c}{N} \Delta W_c^{(t)}, \quad N = \sum_c n_c. \quad (4)$$

5.2 Stage 2: Client-Specific Entropy Binning

Each client runs the global model on its validation set and records the Shannon entropy of the softmax distribution for each example:

$$H(x) = - \sum_{y \in Y} p_y \log p_y. \quad (5)$$

Let $\eta_c = \text{median}_{x \in \mathcal{D}_c^{\text{val}}} H(x)$. Each calibration example is placed in a bin:

$$b_i = \begin{cases} 0 \text{ (easy)} & H(x_i) \leq \eta_c, \\ 1 \text{ (hard)} & H(x_i) > \eta_c. \end{cases} \quad (6)$$

Running example. Client 3 has $|\mathcal{D}_3^{\text{cal}}| = 60$ and $\eta_3 = 0.42$. Twenty-four examples fall into Bin 1 (hard). Their nonconformity scores, sorted ascending, are $[0.012, 0.023, \dots, 0.092, 0.147]$.

5.3 Stage 3: Local Quantile Estimation

Within bin b , client c fits a split-conformal quantile from the $n_{c,b}$ scores in that bin:

$$k = \lceil (n_{c,b} + 1)(1 - \alpha) \rceil, \quad \hat{q}_{\text{local}}^{(c,b)} = s_{(k)}. \quad (7)$$

For Client 3, Bin 1: $k = \lceil 25 \times 0.9 \rceil = 23$, so $\hat{q}_{\text{local}}^{(3,1)} = s_{(23)} = 0.092$.

If $n_{c,b} < n_{\min}$ (we use $n_{\min} = 10$, matching MIN_CALIB_PER_BIN in our implementation), client c falls back to its marginal quantile computed over all calibration examples, irrespective of bin.

5.4 Stage 4: Shrinkage toward the Global Quantile

Each client sends its per-bin nonconformity scores to the server. It is not raw data.

The server pools scores across all clients for each bin:

$$S_{\text{global}}^{(b)} = \bigcup_{c=1}^K \{s_i : b_i = b, i \in \mathcal{D}_c^{\text{cal}}\}, \quad N_b = |S_{\text{global}}^{(b)}|, \quad (8)$$

and computes the global quantile for each bin:

$$\hat{q}_{\text{global}}^{(b)} = \text{Quantile}_{1-\alpha}(S_{\text{global}}^{(b)}). \quad (9)$$

The server broadcasts $\hat{q}_{\text{global}}^{(b)}$ to all clients. Each client forms its final per-bin threshold by a convex combination:

$$\lambda_{c,b} = \frac{n_{c,b}}{n_{c,b} + \tau}, \quad \hat{q}_{\text{final}}^{(c,b)} = \lambda_{c,b} \hat{q}_{\text{local}}^{(c,b)} + (1 - \lambda_{c,b}) \hat{q}_{\text{global}}^{(b)}, \quad (10)$$

where $\tau > 0$ is a hyperparameter. Large $n_{c,b}$ gives $\lambda \rightarrow 1$ (local estimate); small $n_{c,b}$ gives $\lambda \rightarrow 0$ (global estimate).

Running example (continued). With $\tau = 20$ and $\hat{q}_{\text{global}}^{(1)} = 0.054$ from $N_1 = 236$ pooled scores:

$$\lambda_{3,1} = \frac{24}{44} = 0.545, \quad \hat{q}_{\text{final}}^{(3,1)} = 0.545 \times 0.092 + 0.455 \times 0.054 = \mathbf{0.075}.$$

5.5 Inference

At test time, client c assigns the example to bin $b^* = \mathbf{1}[H(x) > \eta_c]$ and builds the prediction set:

$$\mathcal{C}(x) = \{y \in Y : 1 - p_y \leq \hat{q}_{\text{final}}^{(c,b^*)}\}. \quad (11)$$

If $|\mathcal{C}(x)| = 1$ the model commits; otherwise it abstains.

Worked example. The model assigns $p_C = 0.94$ to Mitochondria (correct). The nonconformity score is $1 - 0.94 = 0.06$. With $\hat{q}_{\text{final}}^{(3,1)} = 0.075$, only C passes the threshold: $\mathcal{C}(x) = \{C\}$, a correct commit.

6 Experiments

6.1 Setup

Dataset and clients. We evaluate on a federated multiple-choice QA benchmark with four answer options $Y = \{A, B, C, D\}$. Each client c holds disjoint splits: a training set $\mathcal{D}_c^{\text{train}}$, a validation set $\mathcal{D}_c^{\text{val}}$ used for entropy binning, a calibration set $\mathcal{D}_c^{\text{cal}}$ used for quantile estimation, and a held-out test set $\mathcal{D}_c^{\text{test}}$.

Model. We fine-tune **Qwen2.5-VL-3B-Instruct** [Team, 2025] with LoRA ($r = 8$, $\alpha_{\text{LoRA}} = 16$) for three FedAvg communication rounds with three local epochs per round. The optimiser is AdamW ($\text{lr} = 2 \times 10^{-4}$) with cosine annealing. The global model $f_{\hat{\theta}}$ produces a softmax distribution (p_A, p_B, p_C, p_D) for each example; the LAC nonconformity score is $s_i = 1 - p_{y_i^*}$, where y_i^* is the gold label. We set miscoverage $\alpha = 0.10$, targeting 90% marginal coverage.

Entropy binning. Each client computes the median entropy η_c over its validation set and assigns calibration examples to Bin 0 (easy, $H(x) \leq \eta_c$) or Bin 1 (hard, $H(x) > \eta_c$). Bins with fewer than $n_{\text{min}} = 10$ calibration examples fall back to the client’s marginal quantile.

Baselines. We compare Fed-BLEND against four baselines. **Global-LAC** pools all calibration scores into a single global quantile. **Local-LAC** uses each client’s own marginal quantile with no binning. **Local-APS** replaces the LAC score with the cumulative APS score [Romano et al., 2020]. **Fed-DSCP** applies per-bin local quantiles without shrinkage ($\lambda_{c,b} = 1$).

Algorithm 1 Fed-BLEND: Federated Binned Local-External Nonconformity Distribution

Require: Global model $f_{\hat{\theta}}$; clients $\{c\}_{c=1}^K$; calibration sets $\{\mathcal{D}_c^{\text{cal}}\}$; validation sets $\{\mathcal{D}_c^{\text{val}}\}$; miscoverage α ; shrinkage τ .

Ensure: Per-client, per-bin thresholds $\{\hat{q}_{\text{final}}^{(c,b)}\}$.

— Client-side (parallel) —

- 1: **for** each client c **do**
- 2: Infer on $\mathcal{D}_c^{\text{val}}$; compute $H(x)$ for each example.
- 3: $\eta_c \leftarrow \text{median}_{x \in \mathcal{D}_c^{\text{val}}} H(x)$
- 4: **for** each bin $b \in \{0, 1\}$ **do**
- 5: $\mathcal{D}_{c,b}^{\text{cal}} \leftarrow \{(x, y) \in \mathcal{D}_c^{\text{cal}} : \mathbf{1}[H(x) > \eta_c] = b\}$
- 6: Compute scores $s_i = 1 - p_{y_i^*}$ for all i .
- 7: Compute $\hat{q}_{\text{local}}^{(c,b)}$ via Eq. (7).
- 8: **Send** $\{s_i\}$ to server. $\triangleright$ no raw data leaves client
- 9: **end for**
- 10: **end for**

— Server-side —

- 11: **for** each bin $b \in \{0, 1\}$ **do**
- 12: $S_{\text{global}}^{(b)} \leftarrow \bigcup_c \{s_i : b_i = b\}$
- 13: $\hat{q}_{\text{global}}^{(b)} \leftarrow \text{Quantile}_{1-\alpha}(S_{\text{global}}^{(b)})$
- 14: **Broadcast** $\hat{q}_{\text{global}}^{(b)}$ to all clients.
- 15: **end for**

— Client-side (shrinkage) —

- 16: **for** each client c , bin b **do**
- 17: $\lambda_{c,b} \leftarrow n_{c,b} / (n_{c,b} + \tau)$
- 18: $\hat{q}_{\text{final}}^{(c,b)} \leftarrow \lambda_{c,b} \hat{q}_{\text{local}}^{(c,b)} + (1 - \lambda_{c,b}) \hat{q}_{\text{global}}^{(b)}$ (Eq. (10))
- 19: **end for**

— Test time —

- 20: **for** each test example x at client c **do**
- 21: $b^* \leftarrow \mathbf{1}[H(x) > \eta_c]$
- 22: $\mathcal{C}(x) \leftarrow \{y \in Y : 1 - p_y \leq \hat{q}_{\text{final}}^{(c,b^*)}\}$
- 23: **if** $|\mathcal{C}(x)| = 1$ **then** commit
- 24: **else** abstain
- 25: **end if**
- 26: **end for**

Hyperparameter. The shrinkage temperature is fixed at $\tau = 20$ for all clients and bins.

Metrics. We report five metrics computed over all test examples. *Coverage* is the fraction of examples whose gold label is in $\mathcal{C}(x)$, targeting $\geq 1 - \alpha = 0.90$. *Useful rate* is the fraction of all examples where the model commits correctly ($|\mathcal{C}(x)| = 1$ and $y^* \in \mathcal{C}(x)$). *Hallucination rate* is the fraction where the model commits incorrectly. *Average set size* measures abstention rate indirectly; smaller sets indicate more decisive predictions. *Precision@commit* is accuracy conditioned on the model having committed.

6.2 Baseline Hallucination Rate (No Conformal Prediction)

Table 2 reports results with no conformal post-processing or calibration. The model commits to one answer on every example, so hallucination rate equals 1 – top-1 accuracy. Overall accuracy is 86.95%, but 13.05% of all outputs are wrong committed answers. Client 1 reaches 25.27% hallucination.

Table 2: Per-client performance of the federated LoRA VLM with no conformal post-processing.

Client	Test size	Top-1 accuracy	Hallucination rate
Client 0	74	0.8243	0.1757
Client 1	91	0.7473	0.2527
Client 2	178	0.9157	0.0843
Client 3	60	0.9667	0.0333
Client 4	72	0.8750	0.1250
Overall	475	0.8695	0.1305

6.3 Main Results

Table 3 shows aggregate results at $\alpha = 0.10$.

Table 3: Aggregate performance at $\alpha = 0.10$. Coverage ≥ 0.90 is required. ✓ marks the best value among coverage-valid methods.

Method	Coverage	Useful rate	Hallucination	Avg set size	Prec@Commit
Global-LAC	0.8989	0.8063	0.0674	1.1453	0.9229
Local-LAC	0.9263	0.7811	0.0611	1.2505	0.9275
Local-APS	0.9789	0.2421	0.0000	2.2905	1.0000
Fed-DSCP	0.9432	0.7200	0.0358	1.4126	0.9526
Fed-BLEND	0.9347	0.7663	0.0379 ✓	1.3116	0.9529 ✓

Discussion. Global-LAC misses the coverage target ($0.8989 < 0.90$) and is excluded from comparison. Local-APS achieves zero hallucination but commits on only 24.21% of examples, which is too conservative to be useful. Among the four coverage-valid methods, Fed-DSCP reaches the lowest hallucination rate (0.0358), but it abstains on more questions and gives a useful rate of only 0.72. Fed-BLEND commits on more examples than Fed-DSCP while keeping hallucination low (0.0379). The shrinkage step narrows prediction sets for examples where the local bin has little data, converting some two-option abstentions into correct singleton commits. The net result is the highest Precision@Commit among coverage-valid methods (0.9529) and a useful rate of 0.7663 a 71% relative hallucination reduction from the no-Conformal Prediction baseline.

Useful rate decomposition.

$$\text{Useful rate} = \text{Commit rate} \times \text{Precision@Commit}. \quad (12)$$

As a check: Fed-BLEND commits on 80.6% of test questions, and 95.29% of those commits are correct, giving useful rate ≈ 0.766 .

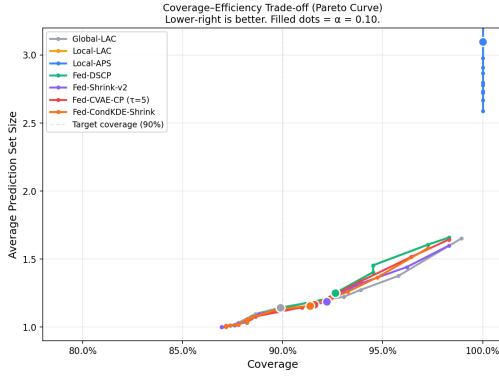
6.4 Client-Wise Analysis

Table 4 breaks down results per client. Fed-BLEND eliminates hallucination entirely on Client 3, the highest-accuracy client, and cuts it most on Clients 0 and 1, which started with the highest hallucination rates.

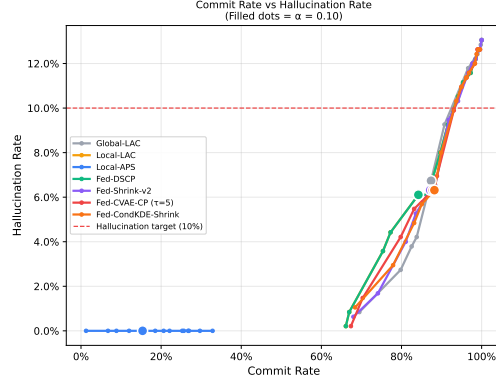
Hallucination vs. commit rate. Figure 2 sweeps α from 0 to 0.30 and plots the resulting Pareto curves. Fed-BLEND and Fed-DSCP both dominate the non-stratified baselines at any fixed commit rate, they hallucinate less. The filled dots mark $\alpha = 0.10$.

Table 4: Per-client results before and after Fed-BLEND.

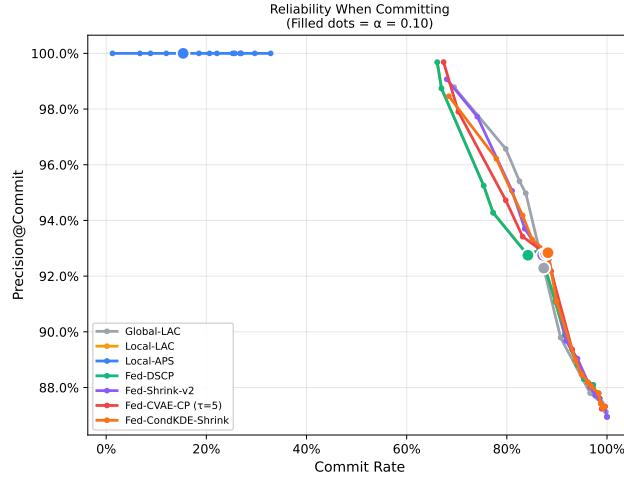
Client	N	Top-1 acc	Hall. (No CP)	Useful (Fed-BLEND)	Hall. (Fed-BLEND)
0	74	0.8243	0.1757	0.7568	0.0541
1	91	0.7473	0.2527	0.5495	0.0440
2	178	0.9157	0.0843	0.8764	0.0449
3	60	0.9667	0.0333	0.8333	0.0000
4	72	0.8750	0.1250	0.7222	0.0278
Overall	475	0.8695	0.1305	0.7663	0.0379



(a) Coverage–efficiency Pareto front. Lower-right is better.



(b) Commit rate vs. hallucination rate. Red dashed line = 10% target.



(c) Precision@Commit vs. commit rate.

Figure 2: Trade-off curves as α is swept from 0 to 0.30. Filled dots mark $\alpha = 0.10$. Additional curves shown in the legend (Fed-CVAE-CP, Fed-CondKDE-Shrink) are ablation variants not discussed in the main paper.

7 Conclusion

We presented **Fed-BLEND**, a post-hoc federated conformal prediction framework for hallucination control in federated fine-tuned VLMs. It stratifies each client’s calibration data by predictive entropy and shrinks the local per-bin quantile toward a global pooled quantile in proportion to local sample size. On a five-client non-IID ScienceQA benchmark, Fed-BLEND reduces hallucinations from 13.05%

to 3.79% a 71% relative reduction, while meeting the 90% coverage requirement and keeping the useful-answer rate at 76.63% with Precision@Commit of 95.29%. It outperforms all four baselines on the safety–efficiency trade-off.

Limitations. Fed-BLEND sends per-bin nonconformity scores scalar statistics, not raw features to the server. A variant using secure aggregation would strengthen the privacy guarantee; we leave this for future work. The coverage guaranty is marginal, not conditional on the difficulty level. Closing the gap between marginal and conditional coverage in the federated setting is an open problem.

Future directions. Four directions stand out. First, extend Fed-BLEND to open-ended VLM outputs using token-level nonconformity scores. Second, replace the plain quantile aggregation with secure multi-party computation. Third, combine Fed-BLEND with RL-based calibration. Fourth, test on larger federated benchmarks such as medical imaging and OpenQnA, etc.

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A Additional Experimental Details

A.1 Hyperparameter Search for τ

We search over $\tau \in \{5, 10, 15, 20, 25, 30, 50\}$. For each value we compute empirical validation coverage and useful rate across all clients. We take the largest τ for which every client reaches coverage ≥ 0.90 . This gives $\tau^* = 20$.

A.2 Conformal Quantile Formula

Given n calibration scores $s_1 \leq \dots \leq s_n$ and miscoverage level α , the conformal quantile is

$$\hat{q} = s_{(k)}, \quad k = \lceil (n+1)(1-\alpha) \rceil. \quad (13)$$

Under exchangeability this satisfies $\Pr(s(X_{\text{test}}, Y_{\text{test}}) \leq \hat{q}) \geq 1 - \alpha$.

A.3 LAC and APS Prediction Sets

LAC.

$$\mathcal{C}_{\text{LAC}}(x) = \{y \in Y : 1 - p_y \leq \hat{q}\}.$$

APS. Sort answers by descending probability: $\pi_1, \pi_2, \dots, \pi_D$ with $p_{\pi_1} \geq p_{\pi_2} \geq \dots$. Let $L_j = \sum_{i=1}^j p_{\pi_i}$. The nonconformity score for true answer y^* at rank r is

$$s_{\text{APS}}(x, y^*) = L_{r-1} + u p_{y^*}, \quad u \sim \text{Uniform}(0, 1).$$

The prediction set includes all answers whose cumulative probability does not exceed $\hat{q}_{\text{APS}}$.

B Uncertainty Summaries

Table 5 lists every scalar summary computed from the softmax distribution. Fed-BLEND uses only `score_correct` (the nonconformity score, Eq. 2) and `entropy` (the stratification signal, Eq. 5). The remaining three quantities (`max_prob`, `margin`, `energy`) are recorded for completeness and used in ablation baselines.

Table 5: Uncertainty summaries from softmax probabilities $\{p_y\}$. Running example: “What is the powerhouse of the cell?” with logits $z_A=1.2$, $z_B=1.8$, $z_C=4.5$, $z_D=0.5$, giving $p_A=0.02$, $p_B=0.03$, $p_C=0.94$, $p_D=0.01$.

Score	Formula	Value	Meaning
score_correct	$1 - p_{y^*}$	0.06	Nonconformity score; low means confident and correct
entropy	$-\sum_y p_y \log p_y$	0.27	Predictive uncertainty; used for easy/hard binning
max_prob	$\max_y p_y$	0.94	Top-class confidence
margin	$p_{\text{top}} - p_{2\text{nd}}$	0.91	Gap between the two most probable classes
energy	$-\log \sum_y e^{z_y}$	-4.68	Free energy; correlates with input hardness

C Prediction-Set Examples

The two examples below trace Fed-BLEND’s inference step on concrete inputs.

Example A: correct commit. Question: “What is the powerhouse of the cell?” Options: A (Ribosome), B (Nucleus), C (Mitochondria, correct), D (Golgi). The model assigns $p_C = 0.94$, giving a nonconformity score of $1 - p_C = 0.06$. Fed-BLEND’s threshold is $\hat{q}_{\text{final}}^{(3,1)} = 0.075$. Only C satisfies $1 - p_C \leq 0.075$, so $\mathcal{C}(x) = \{C\}$: a **correct commit**. For comparison, Fed-DSCP uses $\hat{q} = 0.092$ and also commits to C—the methods agree when the model is confident.

Example B: hallucination prevented. Ground truth: A. Probabilities: $p_A = 0.45$, $p_B = 0.43$, $p_C = 0.07$, $p_D = 0.05$. Nonconformity scores: 0.55 (A), 0.57 (B), 0.93 (C), 0.95 (D). Both A and B have scores below 0.092 (Fed-DSCP’s threshold), so the set is $\{A, B\}$ and the model **abstains**—hallucination avoided. When Fed-BLEND’s shrinkage produces a tighter $\hat{q}_{\text{final}} < \hat{q}_{\text{local}}$, only A passes, converting the abstention into a **correct commit**. This is the mechanism behind Fed-BLEND’s higher useful rate relative to Fed-DSCP.